

15 RAG Terms That You Can't Miss

Agentic RAG

Hallucination

Embeddings

Retrieve

Vector DB

Retriever

Re-ranker

chunking

Grounding

Context Window

Hybrid Search

Metadata Filtering

Similarity Search

Prompt Injection

Latency

Retrieval

The process of fetching relevant documents or passages from an external knowledge base in response to a user query. This is the "R" in RAG without it, the model only knows what it was trained on.

Embeddings

Numerical vector representations of text that capture semantic meaning. Similar sentences have embeddings that are "close" in vector space. The quality of your embeddings directly determines retrieval quality.

Vector Databases

Specialized databases (like Pinecone, Weaviate, Qdrant) built to store and search embeddings at scale. They enable fast nearest-neighbor lookups that power semantic retrieval in RAG systems.



Retriever

The component responsible for fetching relevant chunks from the knowledge base. Can be dense (embedding-based), sparse (BM25/keyword), or hybrid and is one of the most impactful levers in a RAG pipeline.

Chunking

Splitting large documents into smaller, discrete pieces before embedding and indexing. Chunk size and overlap are critical too large loses precision, too small loses context. Often the #1 tuning knob in RAG.

Context Window

The maximum amount of text an LLM can process at once. RAG must carefully fit retrieved chunks into this limit. A larger context window allows more retrieved content, but managing relevance within it is still key.



Grounding

Anchoring the LLM's response to specific, retrieved source documents rather than relying on parametric memory. Grounding improves factual accuracy and makes responses traceable and verifiable.

Re-ranking

A post-retrieval step where a more powerful cross-encoder model re-scores the initially retrieved chunks. Re-ranking surfaces the most genuinely relevant results before passing context to the LLM, boosting answer quality.

Hybrid Search

Combining dense (semantic) and sparse (keyword/BM25) search to maximize retrieval coverage. Semantic search catches meaning; keyword search catches exact matches. Together they handle a much wider range of queries.



Metadata Filtering

Using structured attributes (date, author, category, source) stored alongside embeddings to narrow the search space before vector similarity is computed. Dramatically improves precision for large, heterogeneous corpora.

Similarity Search

The act of finding the closest embeddings to a query vector in the vector database typically using cosine similarity or dot product distance. This is the engine that makes semantic retrieval possible.

Prompt Injection

A security vulnerability where malicious instructions hidden in retrieved documents hijack the LLM's behavior. In RAG systems, attackers can embed adversarial text in indexed content to override system instructions.



Hallucination

When an LLM generates plausible but factually incorrect information. RAG is specifically designed to reduce hallucinations by grounding responses in real retrieved content though it doesn't eliminate the risk entirely.

Agentic RAG

An advanced paradigm where the LLM autonomously decides when, what, and how to retrieve issuing multiple queries, refining results, and orchestrating tools iteratively. Goes beyond one-shot retrieval to multi-step reasoning.

Latency

The total time from query to response, including embedding the query, vector search, re-ranking, and LLM generation. Latency is a critical production concern each step in the pipeline adds overhead that must be carefully managed.

